

Week 5

$$\begin{cases} 3x - 4y = 3 \\ 4x - 5y = 5 \end{cases} \quad \begin{array}{l} \#eq \ 2 \\ \# \text{ unknowns} = \\ \Rightarrow \text{only works} \\ m = n \end{array}$$

• rewrite as a matrix equation

$$\begin{array}{c} A \cdot \bar{X} = \bar{b} \\ \uparrow \\ \text{matrix} \\ \text{multiplication} \end{array}$$

$$\dim(A) = m \times n \rightarrow 2 \times 2$$

$$\dim(\bar{X}) = n \times 1 \rightarrow 2 \times 1$$

$$\dim(\bar{b}) = m \times 1 \rightarrow 2 \times 1$$

$$A = \begin{bmatrix} 3 & -4 \\ 4 & -5 \end{bmatrix}, \quad \bar{X} = \begin{bmatrix} x \\ y \end{bmatrix}, \quad \bar{b} = \begin{bmatrix} 3 \\ 5 \end{bmatrix}$$

$$\hookrightarrow \underbrace{A \cdot \bar{X}}_{\text{LHS}} = \bar{b} \iff \text{same as original S.L.E}$$

- Imagine that  $A$  is invertible matrix

$$\Rightarrow A^{-1} \text{ exists} \quad \leftarrow \text{condition II}$$

$\Rightarrow$  if either or both conditions broken  $\Rightarrow$  must use Augmented matrix

$$\bullet \quad \underbrace{A^{-1} \cdot A \cdot \vec{x}}_{\textcircled{1}} = A^{-1} \cdot \vec{b} \quad \left\{ \begin{array}{l} (-5)3 = (-5)3 \end{array} \right.$$

$$\downarrow$$

$$I_2 \cdot \vec{x}$$

$$\textcircled{2}$$

$$\boxed{\vec{x} = A^{-1} \vec{b}}$$

$\nearrow$  if we know  $A^{-1}$

$$\Rightarrow A = \begin{bmatrix} 3 & -4 \\ 4 & -5 \end{bmatrix} \rightarrow \underline{\underline{A^{-1}}}$$

$$\det(A) = 3 \cdot (-5) - (4)(-4) = -15 + 16 = 1 \neq 0$$

$\hookrightarrow A^{-1}$  exists

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \rightarrow A^{-1} = \frac{1}{\det(A)} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

$$= \frac{1}{\textcircled{1}} \begin{bmatrix} -5 & 4 \\ -4 & 3 \end{bmatrix} = \underbrace{\begin{bmatrix} -5 & 4 \\ -4 & 3 \end{bmatrix}}_{A^{-1}}$$

$\swarrow$   
det

$$\cdot \bar{x} = A^{-1} \cdot \bar{b} = \begin{bmatrix} -5 & 4 \\ -4 & 3 \end{bmatrix} \begin{bmatrix} 3 \\ 5 \end{bmatrix} = \begin{bmatrix} -5 \\ -4 \end{bmatrix} \cdot 3 + \begin{bmatrix} 4 \\ 3 \end{bmatrix} \cdot 5 = \begin{bmatrix} -15 \\ -12 \end{bmatrix} + \begin{bmatrix} 20 \\ 15 \end{bmatrix} = \begin{bmatrix} 5 \\ 3 \end{bmatrix}$$

$$\hookrightarrow \bar{x} = \begin{bmatrix} 5 \\ 3 \end{bmatrix} \Rightarrow (x, y) = (5, 3)$$

# A little theory (formulas)

↳ •  $A, B$       $\dim(A) = \dim(B) = n \times n$

$\Rightarrow A^{-1}, B^{-1} \rightarrow A, B$  are invertible

$\Rightarrow Q$ : how do you compute  $(A \cdot B)^{-1}$ ? if you know  $A^{-1}$  and  $B^{-1}$

A:  $A \cdot B$  is invertible as well

$$(A \cdot B)^{-1} = B^{-1} \cdot A^{-1}$$

multiply inverse  
in reverse order

Collanally:  $\underbrace{A \cdot A \cdot \dots \cdot A}_{\substack{n \times n \\ k\text{-times}}} = A^k$

$$(A^k)^{-1} = (A^{-1})^k = A^{\text{?}} \cdot A^{\text{?}} \cdot A^{\text{?}} \cdot \dots = A^{\text{?}}$$

Theorem 2

Q: Suppose  $A$ ,  $A^{-1}$  exists

find  $(A^T)^{-1} = ?$

$$A: (A^T)^{-1} = \underbrace{(A^{-1})^T}$$

take the inverse of  $A$   
and compute its transpose

$\Rightarrow$  Taylor series  $\swarrow$  Calc 2  
 $\sin\left(\begin{bmatrix} 1 & 1 \\ 2 & 3 \end{bmatrix}\right)$

$$A = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 3 & 0 & 0 \\ 1 & 3 & 5 & 0 \\ 1 & 3 & 5 & 7 \end{bmatrix}$$

Q: find  $A^{-1}$

↳ Sanity: Square?  $\rightarrow 4 \times 4$

↳ use algorithm from  
Last week

↗ you don't know  $A^{-1}$

(do not use shortcut)

↑  
 $\det(A)$

$$\left[ \begin{array}{cccc|cccc} 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 1 & 3 & 0 & 0 & 0 & 1 & 0 & 0 \\ 1 & 3 & 5 & 0 & 0 & 0 & 1 & 0 \\ 1 & 3 & 5 & 7 & 0 & 0 & 0 & 1 \end{array} \right]$$

ERO on LHS

↳ same on RHS

↳ put LHS  $\rightarrow$  REF

$$\left[ \begin{array}{cccc|cccc} 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 & -1 & 1 & 0 & 0 \\ 0 & 3 & 5 & 0 & -1 & 0 & 1 & 0 \\ 0 & 3 & 5 & 7 & -1 & 0 & 0 & 1 \end{array} \right]$$

$$r_2 \rightarrow r_2 - r_1$$

$$r_3 \rightarrow r_3 - r_1$$

$$r_4 \rightarrow r_4 - r_1$$

← same RHS

$$r_3 \rightarrow r_3 - r_2$$

$$r_4 \rightarrow r_4 - r_2$$



$$\left[ \begin{array}{cccc|cccc} 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 & -1 & 1 & 0 & 0 \\ 0 & 0 & 5 & 0 & 0 & -1 & 1 & 0 \\ 0 & 0 & 5 & 7 & 0 & -1 & 0 & 1 \end{array} \right]$$

Skipping  
Step for  
time

Blinked  
and we skipped  
to the end

$$A \cdot \bar{X} = \bar{b}$$

if  $A$  is invertible  $\Rightarrow$  there is a unique solution

$$A^{-1} \cdot (A \cdot \bar{X}) = A^{-1} \cdot \bar{b}$$

$$\underbrace{(A^{-1} \cdot A)}_{I_n} \cdot \bar{X} = A^{-1} \cdot \bar{b}$$

$$\bar{X} = A^{-1} \cdot \bar{b}$$

$$A \cdot B \cdot C = (A \cdot B) \cdot C = A \cdot (B \cdot C)$$

LHS

$$I_n \cdot \bar{X} = \bar{X} = A^{-1} \cdot \bar{b}$$

↑ identity

↑ solution by construction

inverse matrix  
if exists is unique

Ex

$$\begin{cases} x_1 + 2x_2 + 3x_3 = 5 \\ 2x_1 + 5x_2 + 3x_3 = 3 \\ x_1 + 4x_3 = 17 \end{cases}$$

$$[A | I_3] = \left[ \begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 2 & 5 & 3 & 0 & 1 & 0 \\ 1 & 0 & 3 & 0 & 0 & 1 \end{array} \right] \xrightarrow[r_3 \rightarrow r_3 - r_1]{r_2 \rightarrow r_2 - 2r_1} \left[ \begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & -3 & -2 & 1 & 0 \\ 0 & -2 & 5 & -1 & 0 & 1 \end{array} \right] \xrightarrow[r_3 \rightarrow r_3 + 2r_2]{r_1 \rightarrow r_1 - 2r_2} \rightarrow$$

REF

→

Answer

$$A^{-1} = \begin{bmatrix} -40 & 16 & 9 \\ 13 & -5 & -3 \\ 5 & -2 & -1 \end{bmatrix}$$

$$\vec{x} = \begin{bmatrix} 5 \\ 3 \\ 17 \end{bmatrix} \cdot A^{-1} = \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix}$$

solution exists

Lecture 11 is important

$$\text{if : } A^{-1} \Rightarrow \bar{x} = A^{-1} \cdot b$$

no matter  
what  $b$

↗ solution exists  
↘ unique

$A^{-1}$  doesn't exist  
 $\Rightarrow$  solution for some  $\bar{b}$

$$\begin{bmatrix} 1 & 1 & 2 & b_1 \\ 1 & 0 & 1 & b_2 \\ 2 & 1 & 3 & b_3 \end{bmatrix} \rightarrow \begin{matrix} r_2 \rightarrow r_2 - r_1 \\ r_3 \rightarrow r_3 - 2r_1 \end{matrix} \rightarrow \begin{bmatrix} 1 & 1 & 2 & b_1 \\ 0 & -1 & -1 & b_2 - b_1 \\ 0 & -1 & -1 & b_3 - 2b_1 \end{bmatrix}$$

$$\rightarrow \begin{matrix} r_3 \rightarrow r_3 - r_2 \\ r_2 \rightarrow -r_2 \end{matrix} \rightarrow \begin{bmatrix} \textcircled{1} & 1 & 2 & b_1 \\ 0 & \textcircled{1} & 1 & b_1 - b_2 \\ 0 & 0 & 0 & b_3 - b_1 - b_2 \end{bmatrix} \Leftarrow \text{REF-like}$$

focus on  $r_3$ : system is consistent only if

$$b_3 - b_1 - b_2 = 0$$

$$b_3 - b_1 - b_2 = 0$$

$$\begin{bmatrix} 1 & 1 & 2 & b_1 \\ 0 & 1 & 1 & b_1 - b_2 \\ 0 & 0 & 0 & 0 \end{bmatrix} \rightarrow r_1 \rightarrow r_1 - r_2 \rightarrow \begin{bmatrix} \textcircled{1} & 0 & 1 & b_2 \\ 0 & \textcircled{1} & 1 & b_1 - b_2 \\ 0 & 0 & 0 & 0 \end{bmatrix} \Leftarrow \text{RREF}$$

there are infinitely many solutions for any

$$b = \begin{bmatrix} b_1 \\ b_2 \\ b_3 = b_1 + b_2 \end{bmatrix}$$

$$\vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} b_2 - s \\ b_1 - b_2 - s \\ s \end{bmatrix}$$

however  $b_3 - b_1 - b_2 \neq 0$

is inconsistent (No solutions)

Can be anything

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix} \quad B = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$

diagonal matrix

Note:

$$A^2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 9 \end{bmatrix}$$

Can multiply both ways in diagonal matrix